ART801

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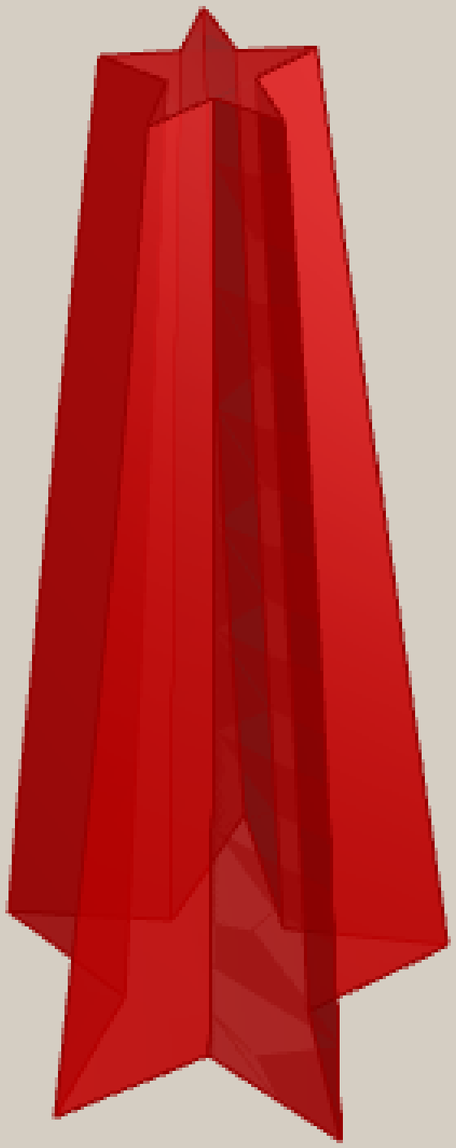
Introduction to 3D Modelling & Parametric Design

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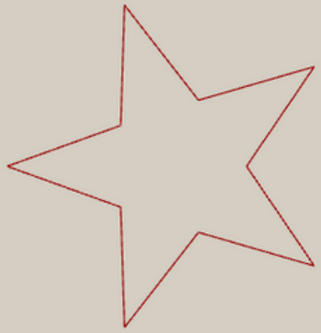
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1.1 Star Tower

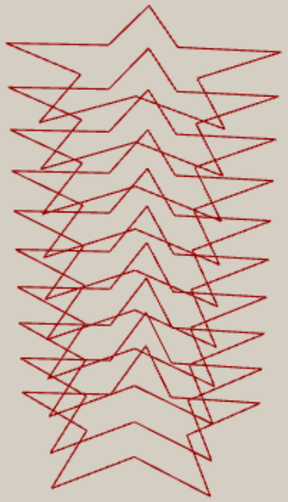
This model is created using the Lunchbox plugin. LunchBox is a versatile plug-in for Grasshopper, the visual programming language linked with Rhinoceros 3D, that improves parametric modelling features. LunchBox has components for creating various panelling systems, including quad grids, diamonds, and triangles. The tool's flexibility enables quick iteration and refinement of panelling concepts, making it beneficial in both the early stages of design exploration and the more detailed development phases.







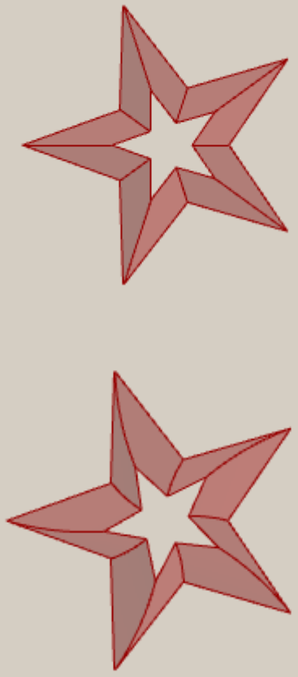
1. Create a closed curve for the tower.



2. Create a linear array of the curve in the Z direction.



3. Scale the arrays and construct a domain to define boundaries on how much they were scaled.



4. Rotate the arrays using another domain to define their inclination and create a loft..

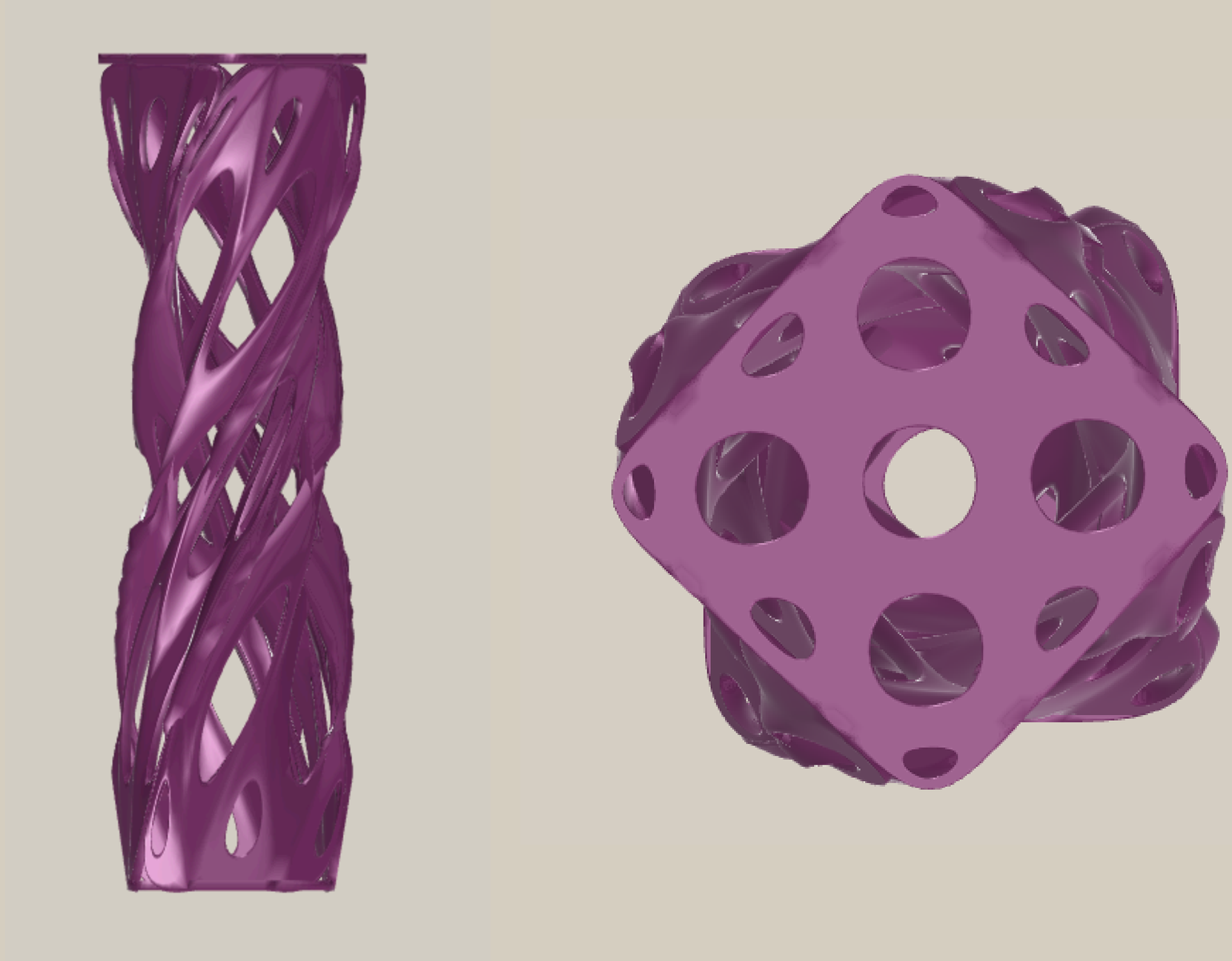


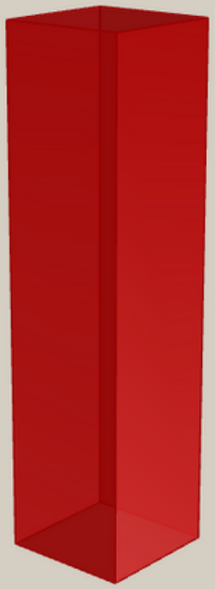


5. Finally, use the Cap Holes command to create the tower.

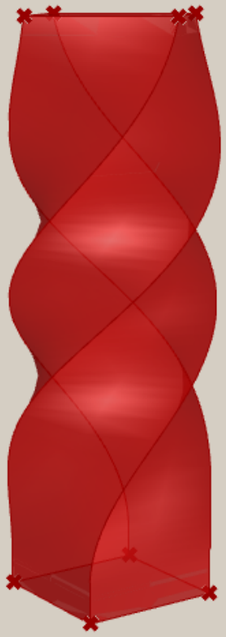
1.1 Weaverbird Structure

WeaverBird is a specialized plugin for Grasshopper. It focuses on topological mesh manipulation, subdivision, and mesh transformations inside Grasshopper. WeaverBird is often used in combination with other Grasshopper plugins to create complex mesh-based designs and patterns, enhancing Grasshopper's mesh editing capabilities for parametric and algorithmic design approaches.





1. Start by creating a rectangle and extruding it to create a tower.



2. Then by creating a line in the tower's middle, twist it by its axis.



3. Using the Deconstruct Brep command, create panels on the tower's surfaces.



4. Next, create a Simple Mesh to the object, and then a Mesh Join to connect the surfaces into one



5. At this stage, incorporate Weaverbird features, such as Picture Frame, which creates a new mesh where each frame has a new hole in the centre.



6. Then, use Weaverbird's Catmull-Clark subdivision command, which is a tool for smoothing out the mesh of the object by adding new vertices and creating new faces.

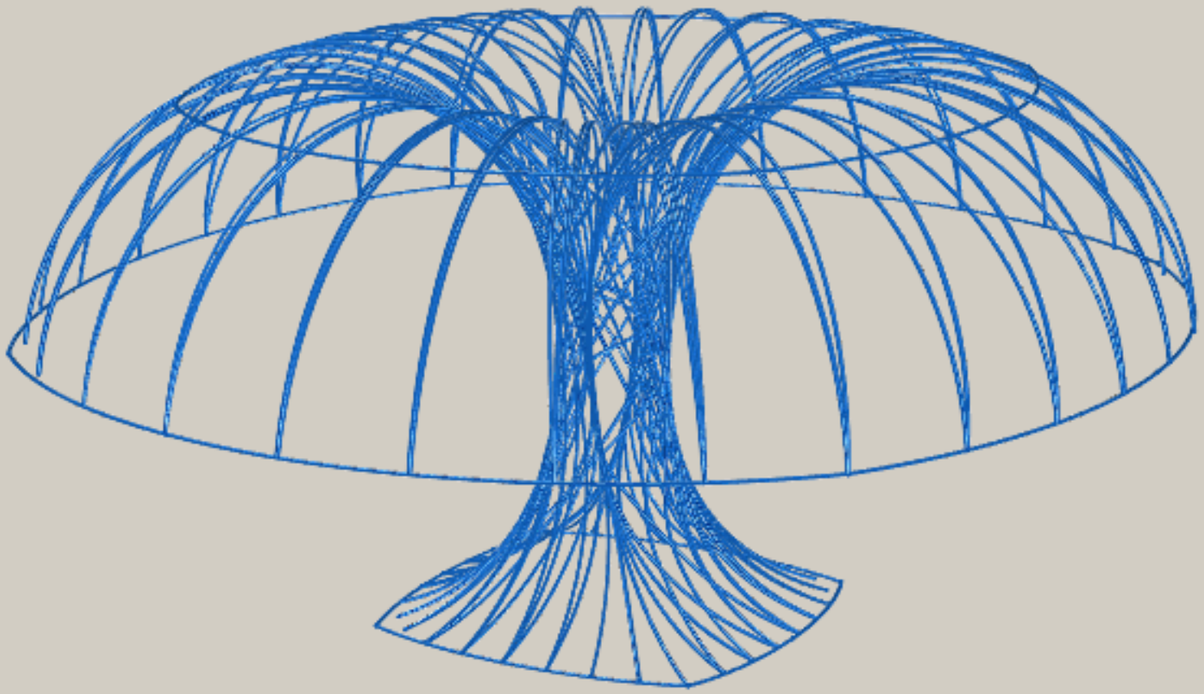


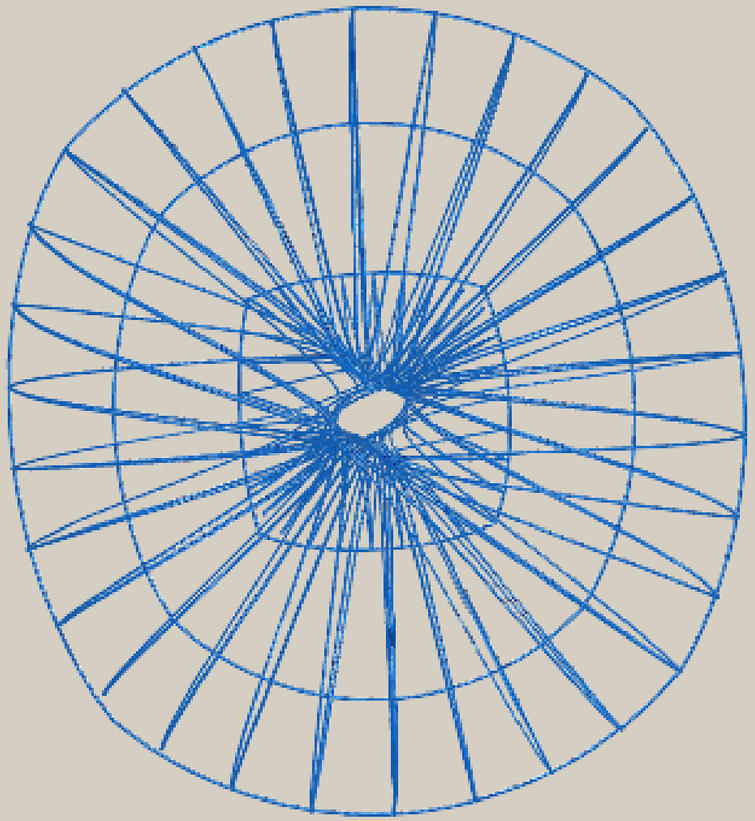
7. Similarly, use the Weaverbird Mesh Thicken command which creates a new mesh that is a closed solid.

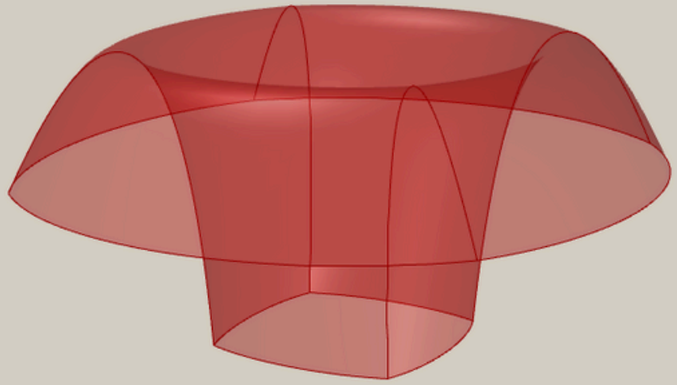


8. Finally, add a List Item to the model which extracts different surfaces from the model, and then use Picture Frame and Weaverbird Mesh Thicken commands to finalize the tower.

1.3 Tree Lamp



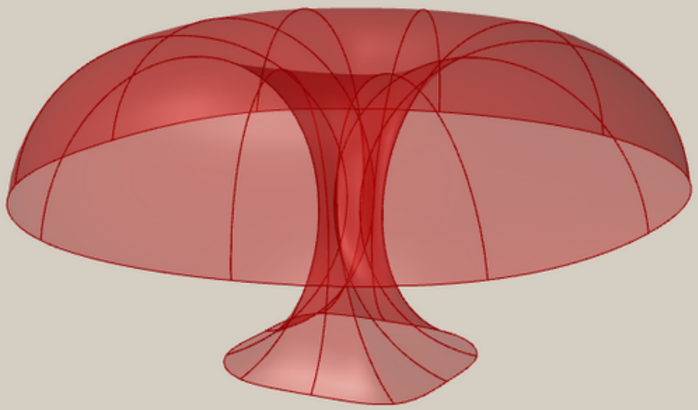




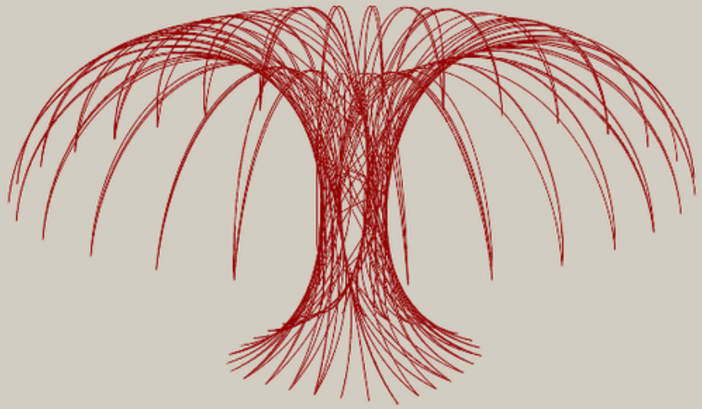
1. Create three curves and create a surface on the curves using the Loft option.



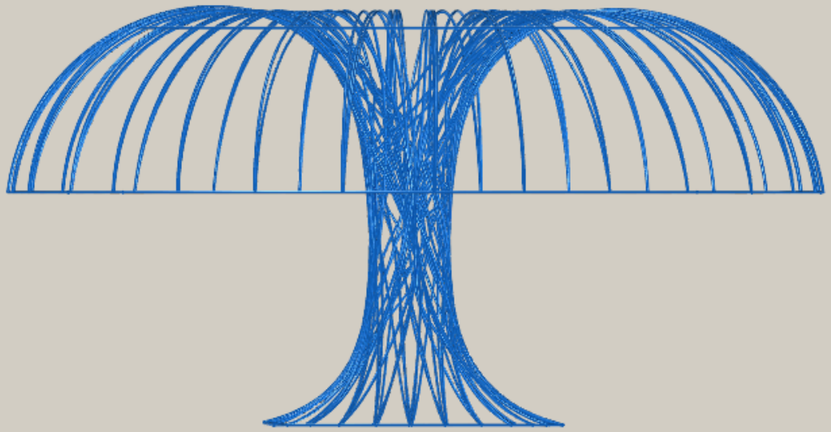
2. Divide the curves into equal lengths and use the Explode Tree option to create “branches” on the surface.



3. Create a 3D Arc on the surface.



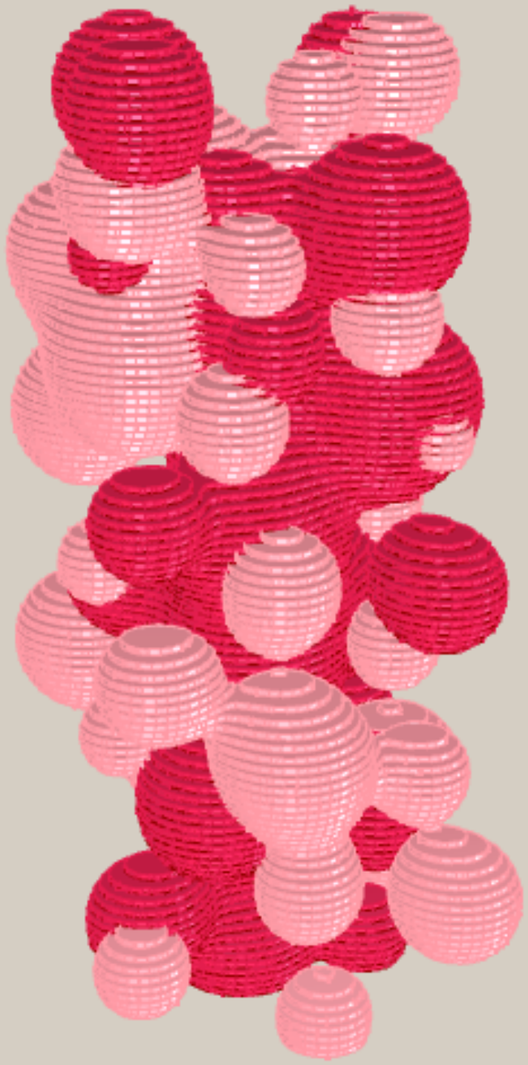
4. Rebuild the curve with a specific number of control points and create a surface geodesic between points.

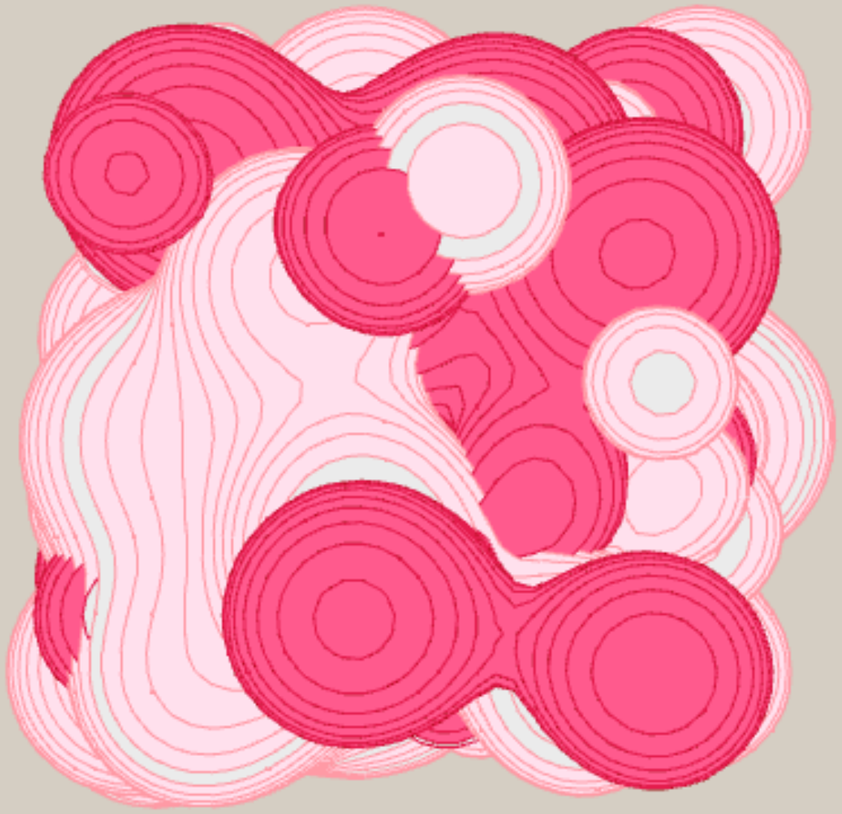


5. Pipe all the curves to give them depth.

1.4 Metaball Sructure

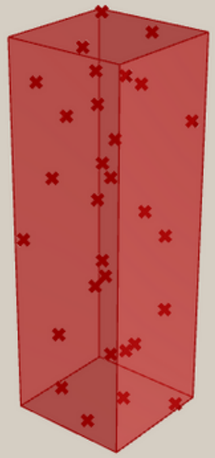
The metaball function in Grasshopper is a strong tool for creating organic, blob-like 2D curves from a series of input points and their charge values. It uses a field summation technique to generate smooth, continuous shapes that blend and merge as the points become closer together. The function creates these curves by sampling the space around each point, computing elevations using a fall-off equation, and intersecting the resulting landscape with a horizontal plane at a given iso-surface value. This approach enables designers to build complex, flowing geometries suitable for a wide range of parametric design applications, including architectural forms and digital art.



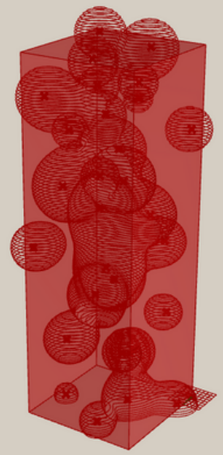




1. Create a 65x65 rectangle.



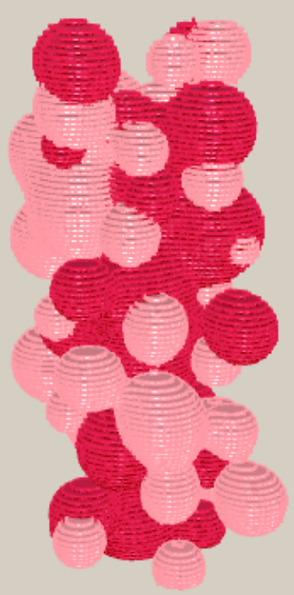
2. Extrude the rectangle and use Populate 3D to make points on the model.



3. Use the Metaball function which creates 2D isocurves based on a set of input points. Generate these points by using a random generator and a domain.



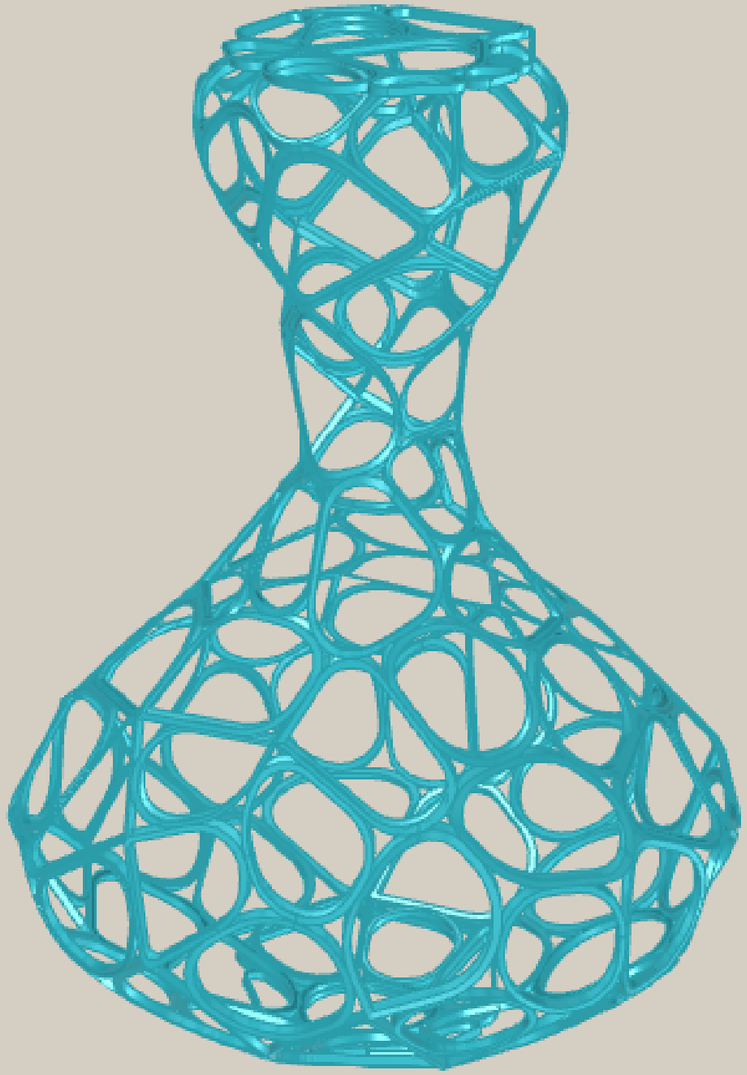
4. Create a Linear Array by going up the model and putting a boundary surface on the metaballs.

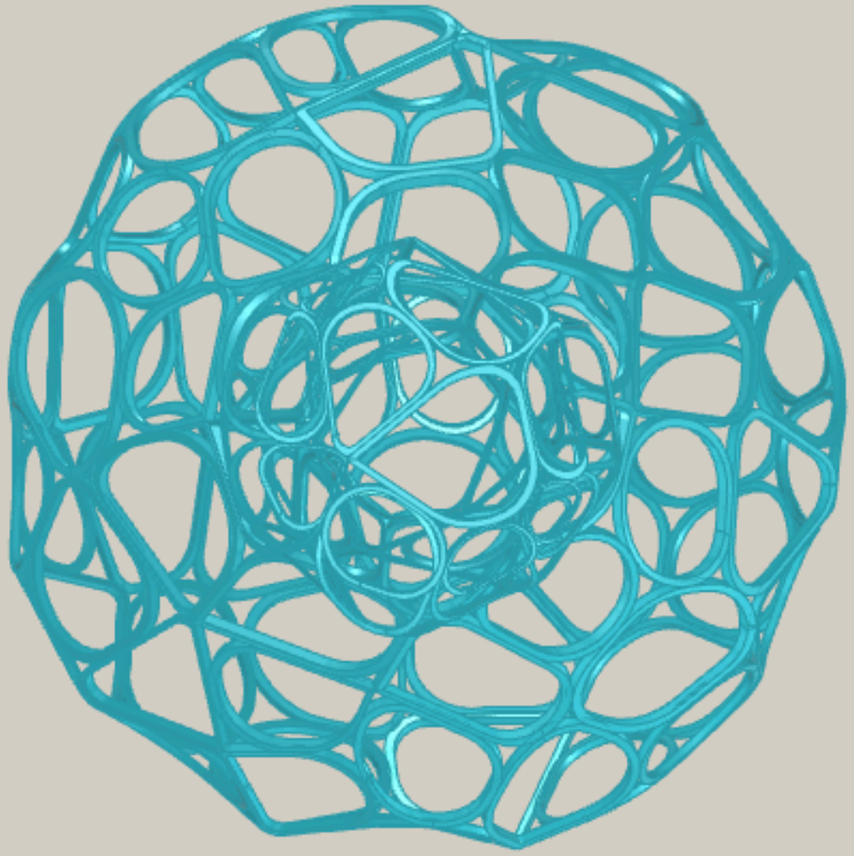


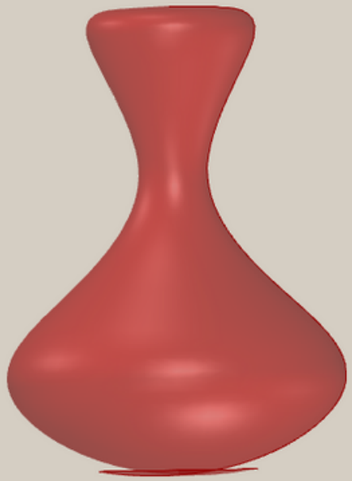
5. Finally extrude the curves along the Z-direction. Create the same workflow for a different set of metaballs creating a different domain. Use the same Extrude option for the metaballs so they intersect each other and form a final pattern.

2.1 Voronoi vase

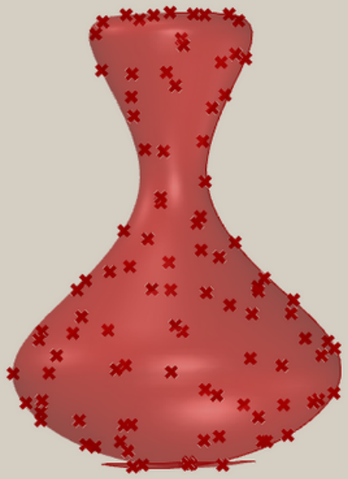
Voronoi diagrams in Grasshopper provide an effective method for creating cellular patterns inspired by nature. Voronoi component requires points, a boundary, and a plane as inputs. To generate points, we use the Populate 2D component, which makes random points within a defined boundary. The resulting Voronoi diagram will display in the Rhino viewport, dividing the space into cells, each containing the area closest to the central point. This basic arrangement allows for easy experimentation with various patterns and forms, making it a useful tool for developing intricate, nature-inspired structures.







1. To start the model create two curves which are in the shape of a half vase. Then using the Revolution command, create a revolution surface to make the full vase.



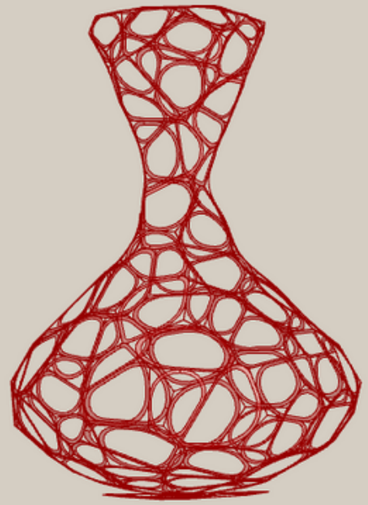
2. Use Populate Geometry command, (which is the key option to create a Voronoi pattern), and sepcifie points on the vase.



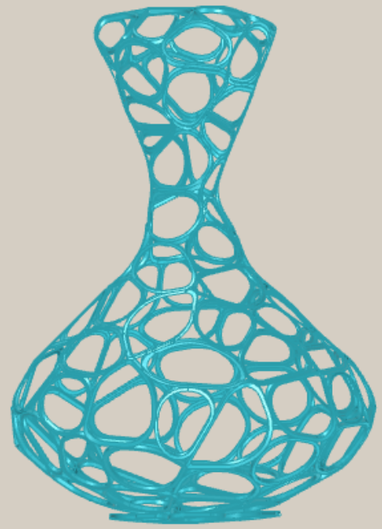
3. Use the Voronoi 3D command to create cellular patterns. Use the Brep command to create boundaries on the vase according to its population points.



4. Use the Polyline command to create lines surrounding the points and scale them to offset the lines.

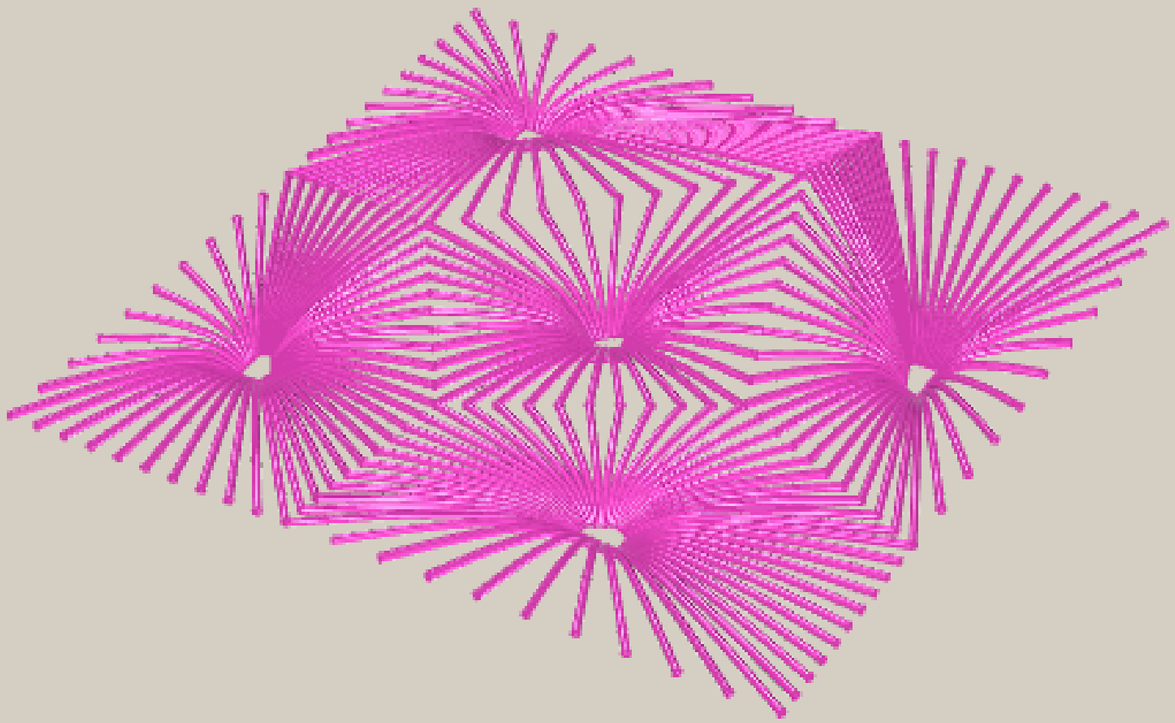


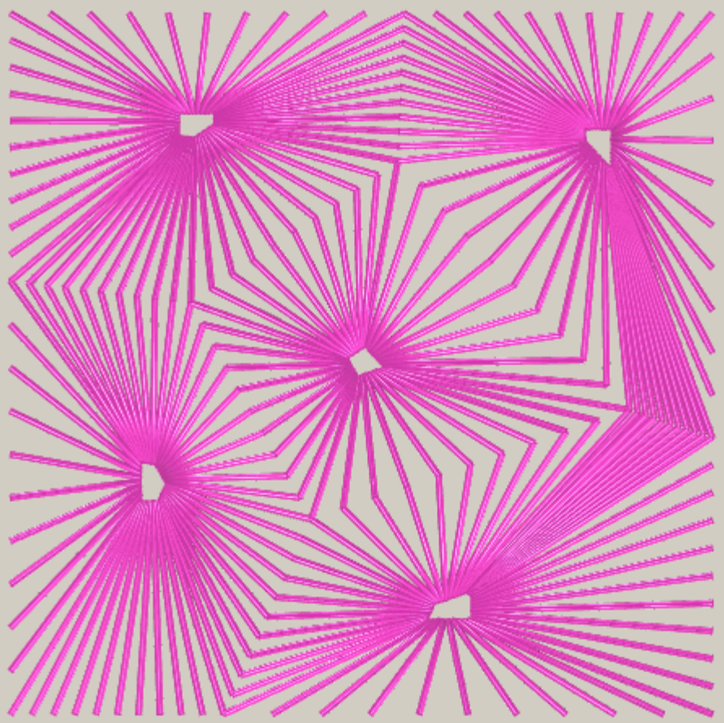
5. Use the Loft command to make a surface on the lines

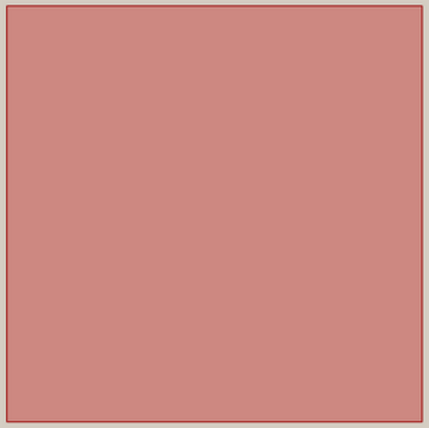


6. Finally, use the Extrude command to give the lines some depth to make a Voronoi vase.

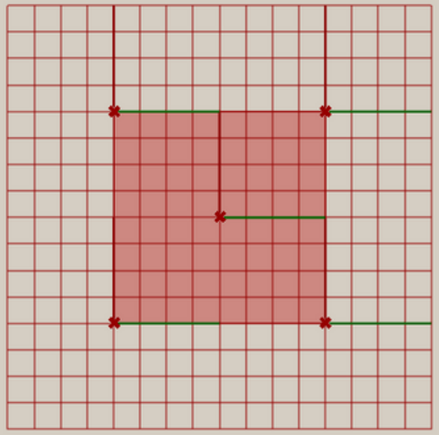
2.2 Voronoi Roof Structure



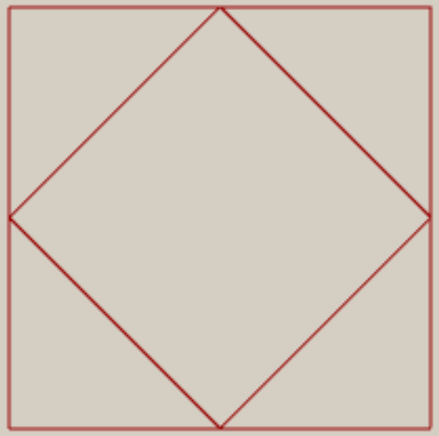




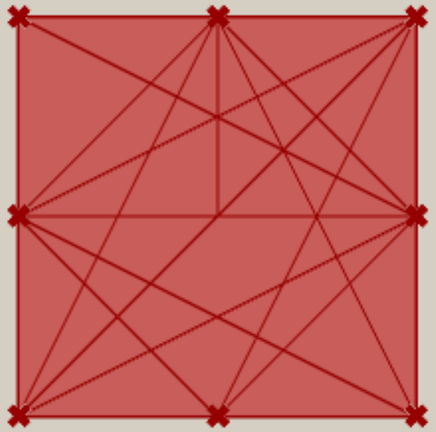
1. Start by creating a rectangular surface.



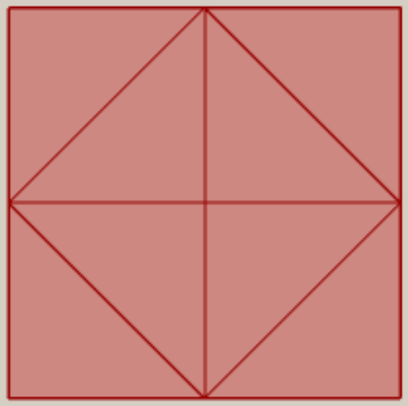
2. Create 4 multidimensional sliders to evaluate surface properties.



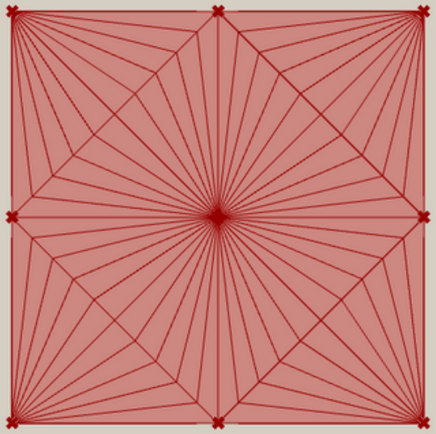
3. Create a Voronoi diagram on the surface.



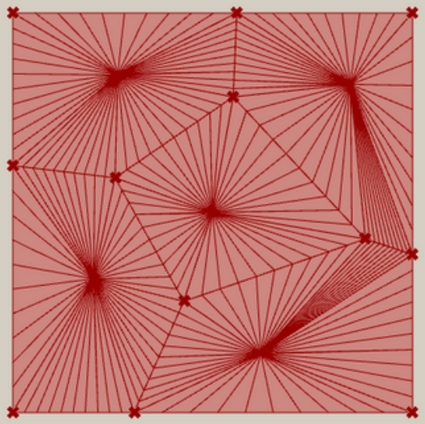
4. Using Deconstruct Brep to split the surface into parts and create lines to connect the centres of the multidimensional sliders.



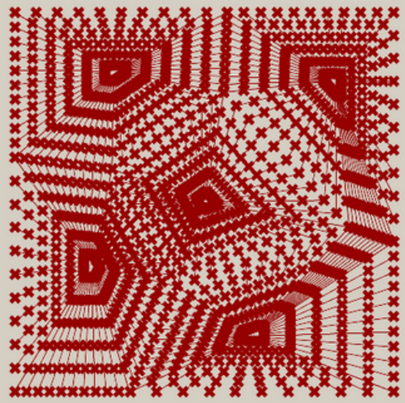
5. Create a point connecting it to the sliders and create a shift lift to move the coordinates of the lines.



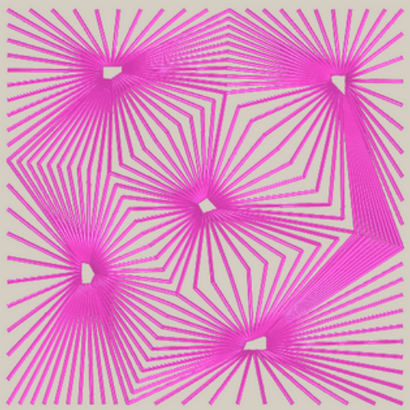
6. Use the Tween Curve command to create curves between points and divide them into equal segments.



7. Play with the multi-dimensional sliders and move them around to change the pattern.



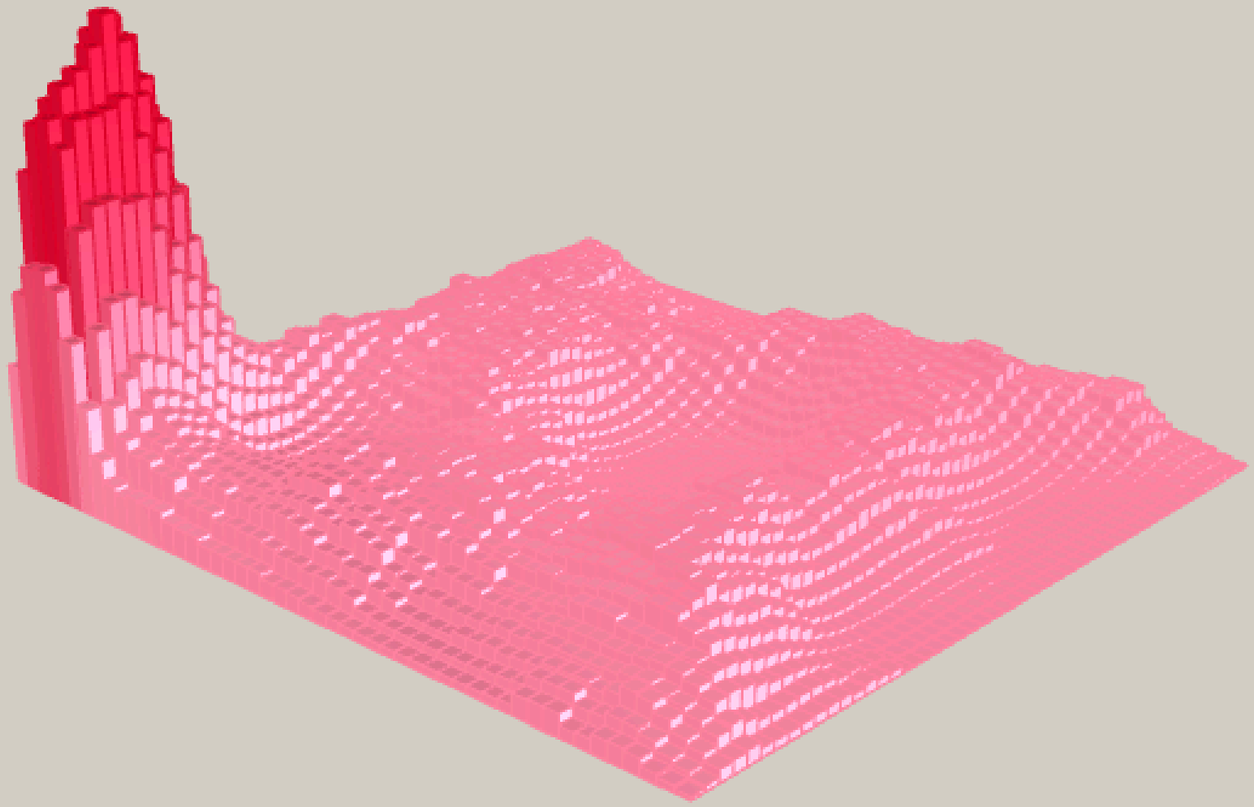
8. Move the surface in the Z direction and interpolate the curves.

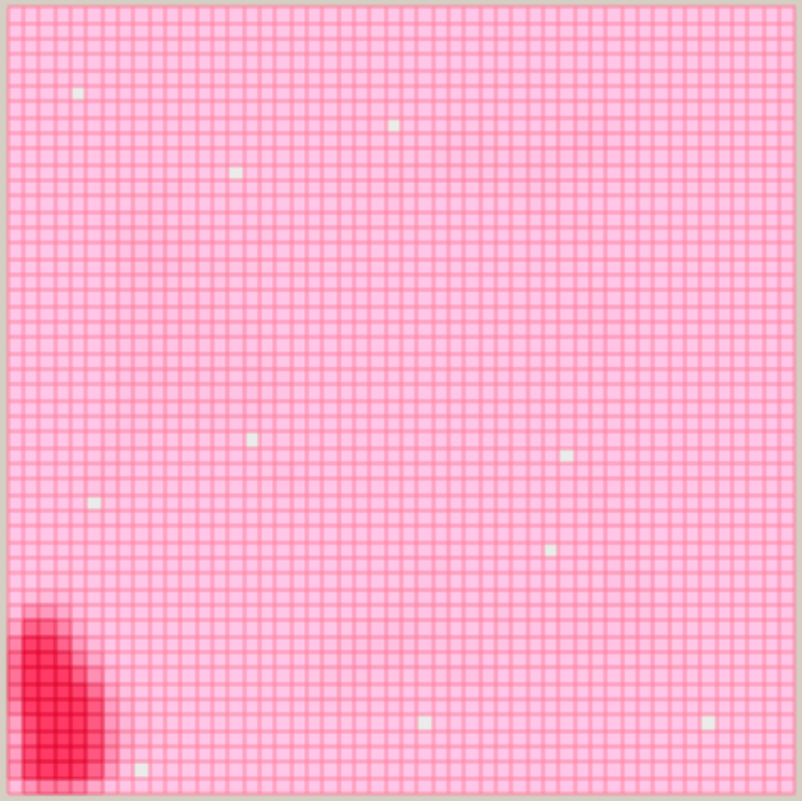


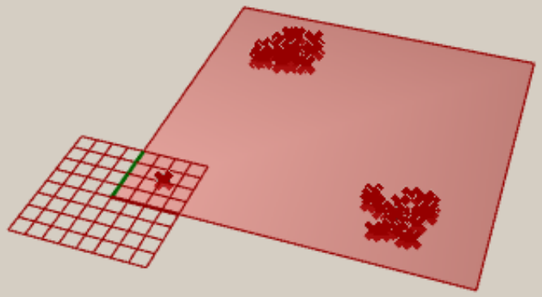
9. Create a pipe around the curves to create the desired roof shape.

3.1 Physorealm Tower

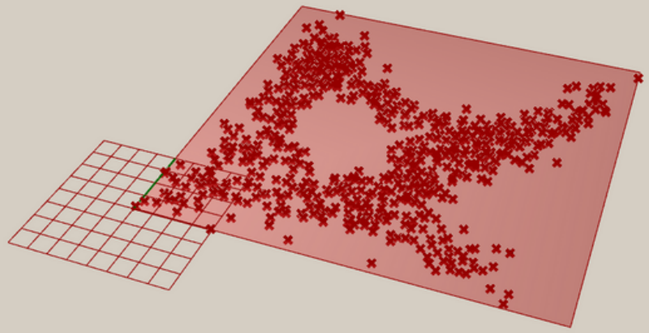
Physarealm is a multi-objective system as a Grasshopper plugin that was inspired by the behaviour of Physarum polycephalum, a slime mould. Ma Yidong created this open-source application, which simulates the development, growth, and minimisation of transportation networks using stigmergy techniques similar to ant colony optimisation. Physarealm enables us to create sophisticated, organic-looking structures in 3D, expanding upon the initial 2D implementations. The plugin supports artistic expression and may simulate slime mould behaviour on meshes. Physarealm's agent-based system uses simple individual behaviours to mimic the biological behaviour of Physarum, making it a powerful tool for investigating bio-inspired design ideas within the Grasshopper environment.



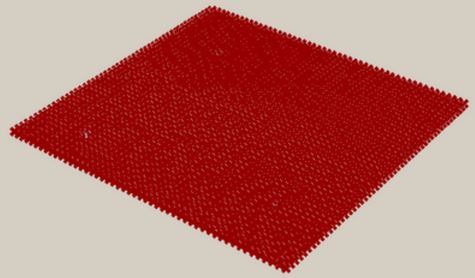




1. Create a 50x50 plane and assign 3 points to it. Create a physarealm and assign 2 emitter points and one food point. Food sources influence the growth and behaviour of the simulated points.



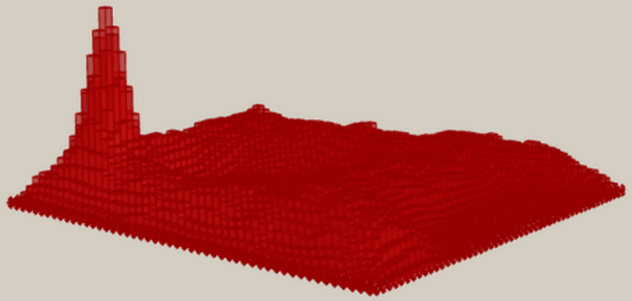
2. Connect a timer to the physarealm and a toggle. The points will start moving in a stigmergic way when the toggle is pressed.

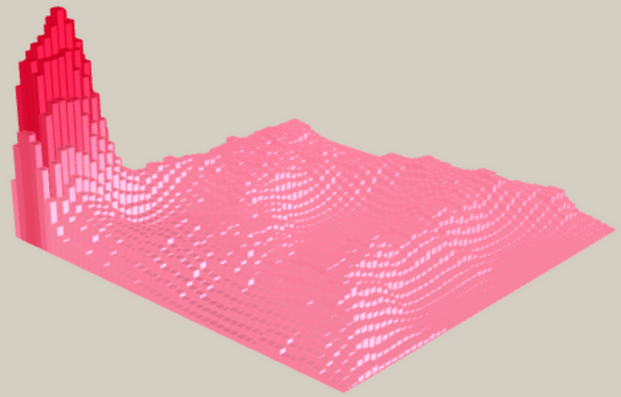


3. Use the Field Points command which gives all points on the surface a chemoattractant value These field points are used to create complex, organic-looking structures and simulate natural growth processes.



4. Create a domain box and connect it to the field points.

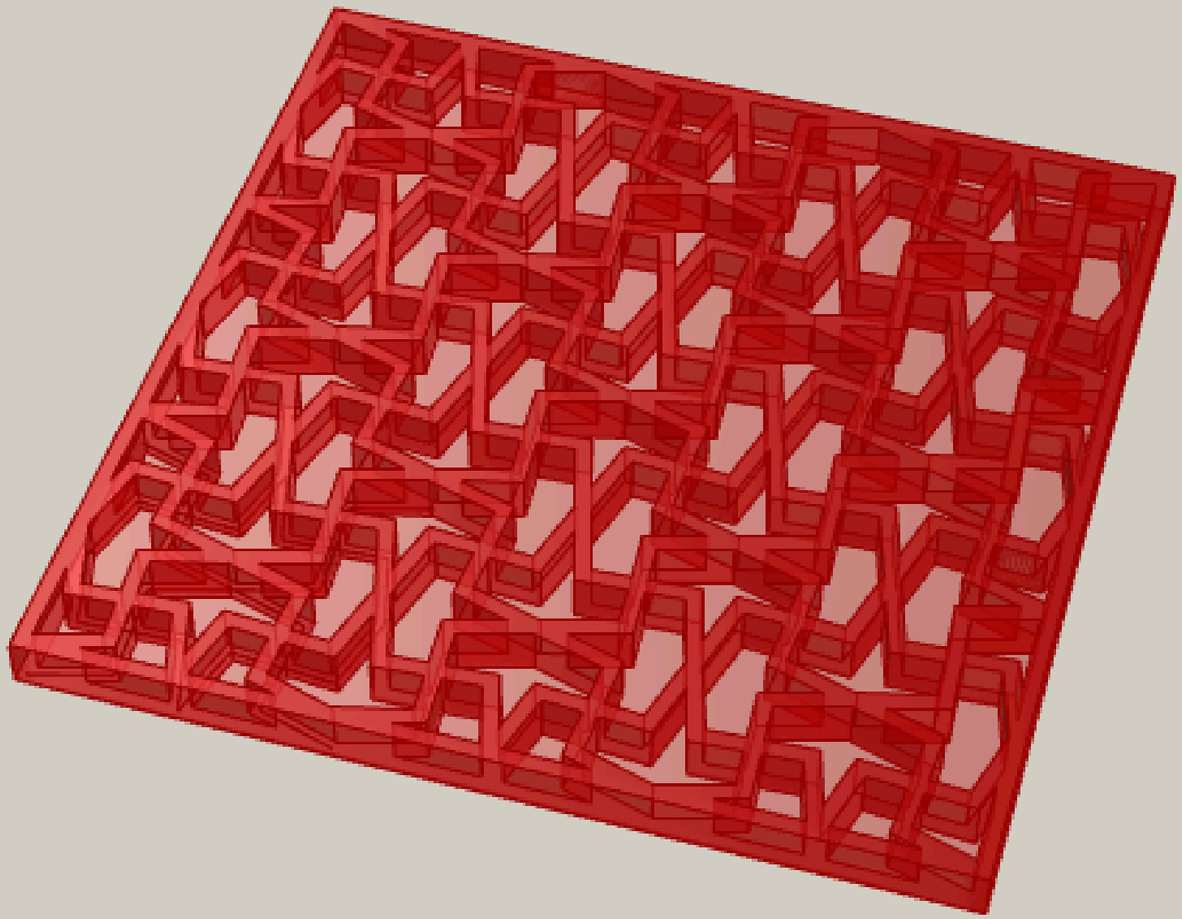


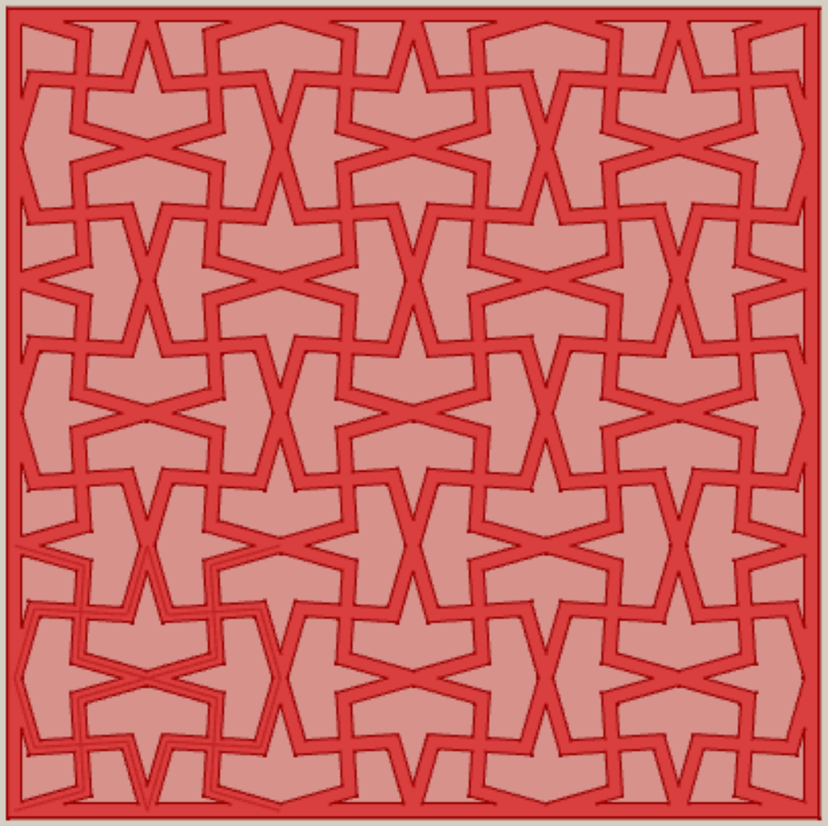


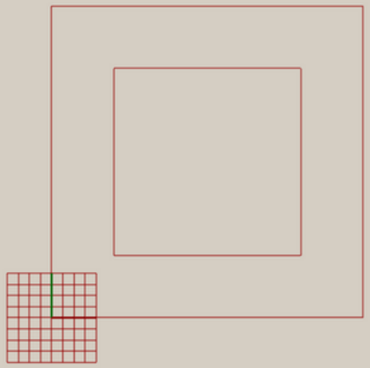
5. Finally setting the timer on again the points start climbing the box up in a stigmeric way.

4.1 Shape grammar

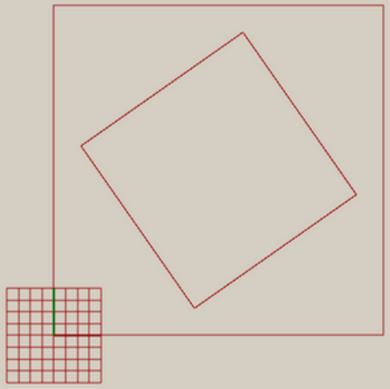
Shape grammars are a framework for generating complicated geometric designs using basic shapes and transformation rules. They are made up of shape rules and a generation engine that identifies and interprets these rules. The following shape is created with the help of Lunchbox, which is a plug-in for Grasshopper.



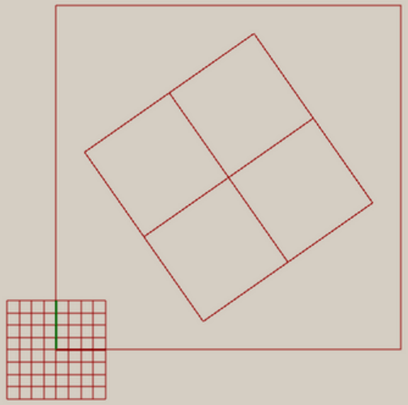




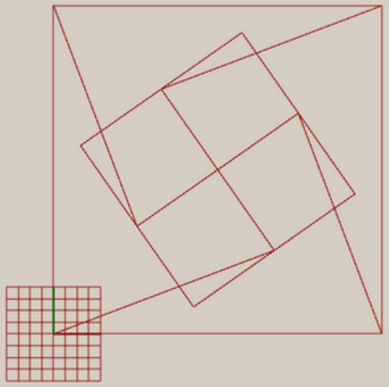
1. Step one is to create a rectangle and use the scale option to create a smaller one in the middle.



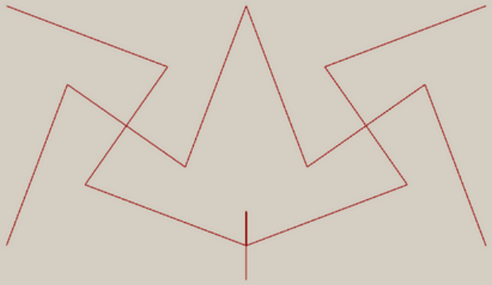
2. The next step is to rotate the smaller rectangle.



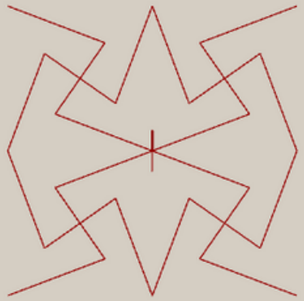
3. Using the explode option, the rectangle gets separated into smaller segments related to the centre.



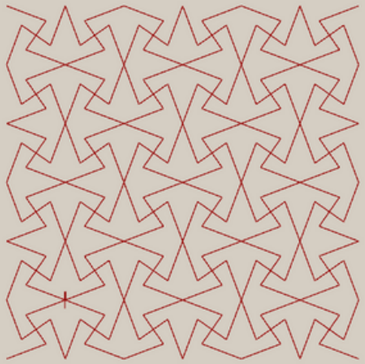
4. To create the desired shape, use the Explode command to connect the big triangle to the smaller one and connect the segments together.



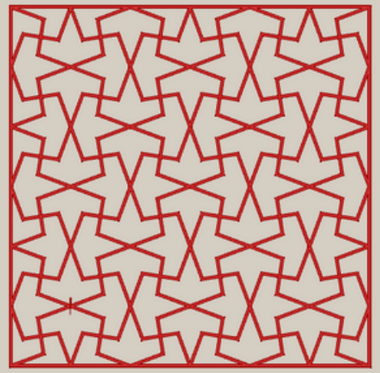
5. After the desired shape is created, mirror the image through the x-axis.



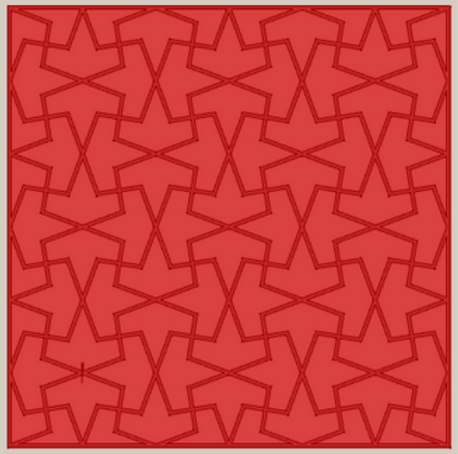
6. Furthermore, mirror the image on the XZ plane.



7. By using a rectangular array, multiply the image creating a pattern.



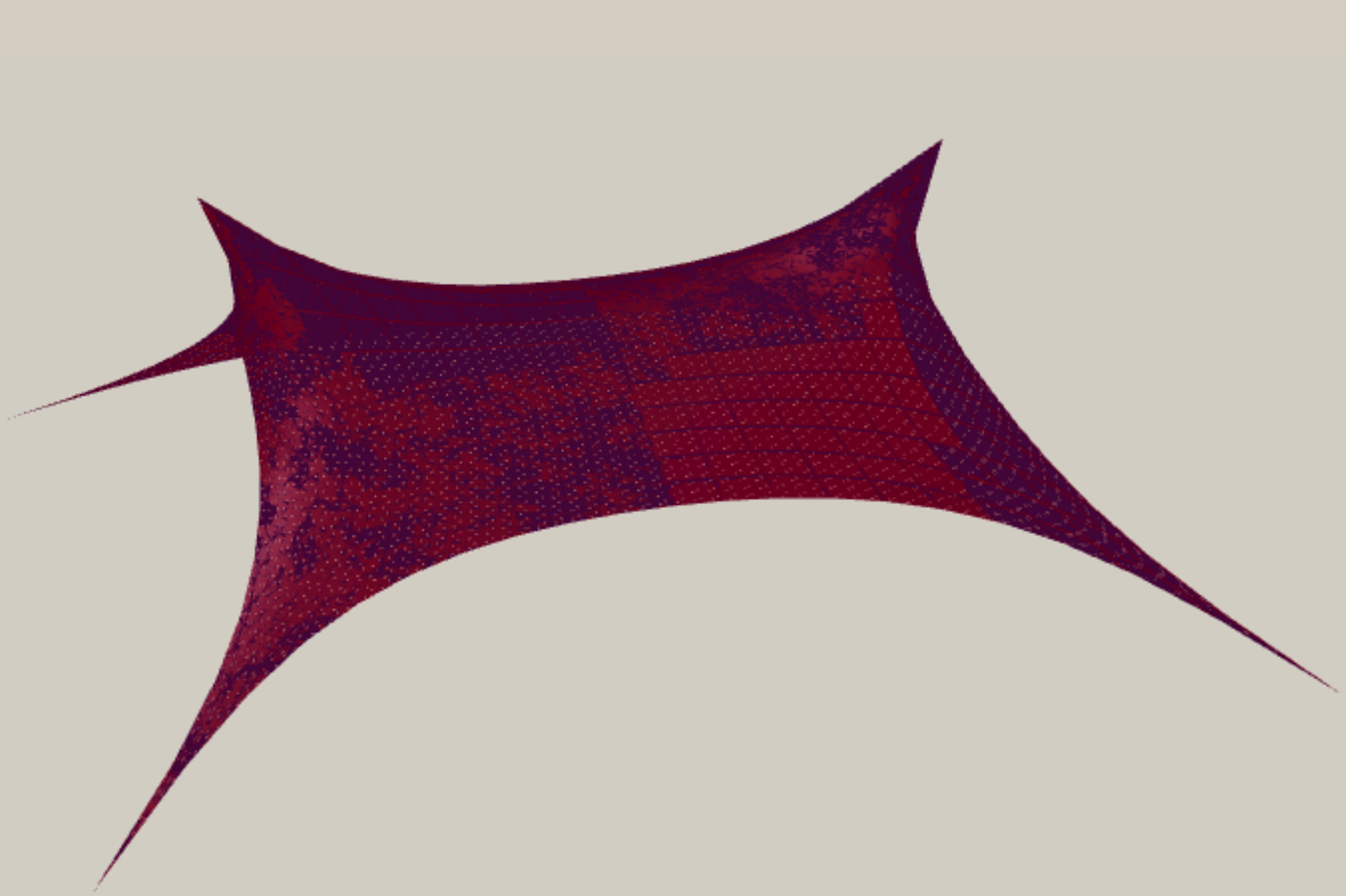
8. Come up with a design by playing with the points on the curves by changing the angle.

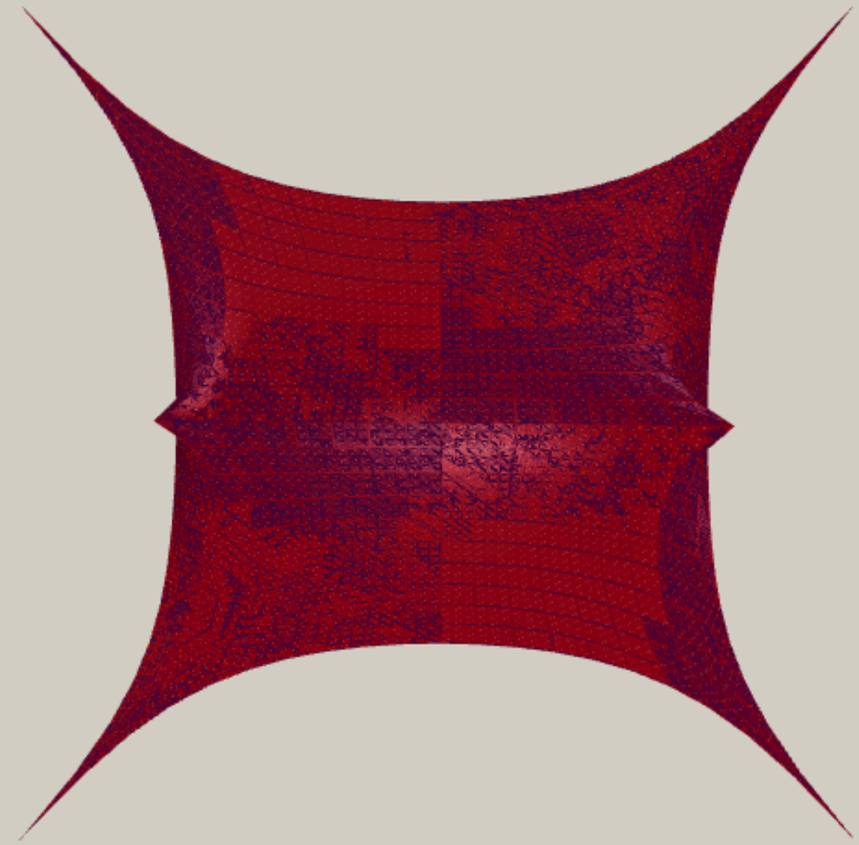


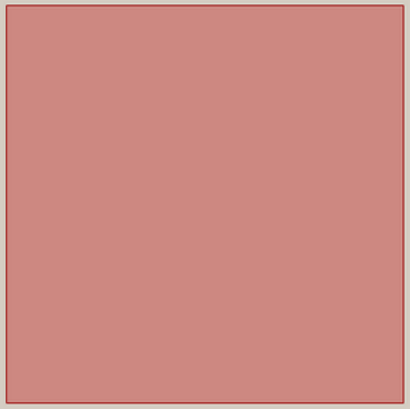
9. Lastly use offset to create a boundary on the lines and give the pattern a surface.

4.2 Kangaroo Tent

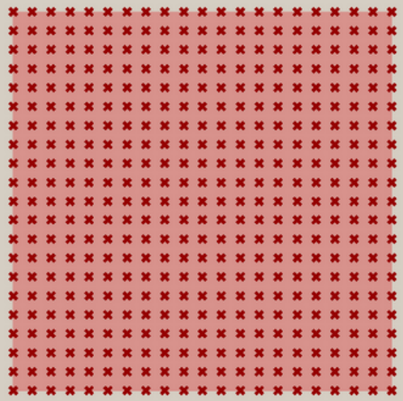
Kangaroo is a strong physics engine plugin for Grasshopper that is built into Rhino. It allows for interactive simulations, form-finding, optimisation, and constraint solution in the parametric design environment. Kangaroo simulates physical behaviours and forces using a particle-spring system and an energy minimisation approach. The plugin includes a solver framework and a set of Grasshopper components that enable users to design complicated simulations by combining various goals such as length constraints, point loads, and geometry constraints. Kangaroo has a wide range of applications, including structural analysis, cloth simulation, and generative design. Its flexibility and connection with Grasshopper make it a great tool for architects, engineers, and designers exploring form-finding and physics-based design solutions.



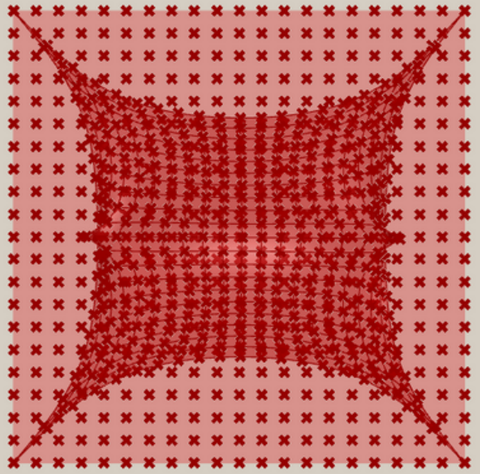




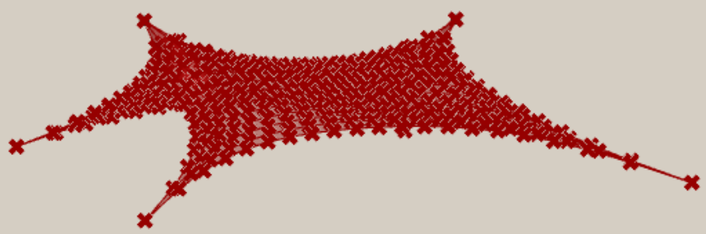
1. Create a 100x100 rectangle.



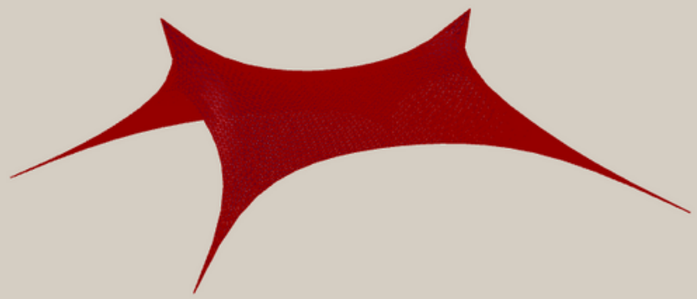
2. Use the Naked Vertices command on Kangaroo to separate the vertices of a mesh into two lists: naked (boundary) vertices and clothed (interior) vertices.



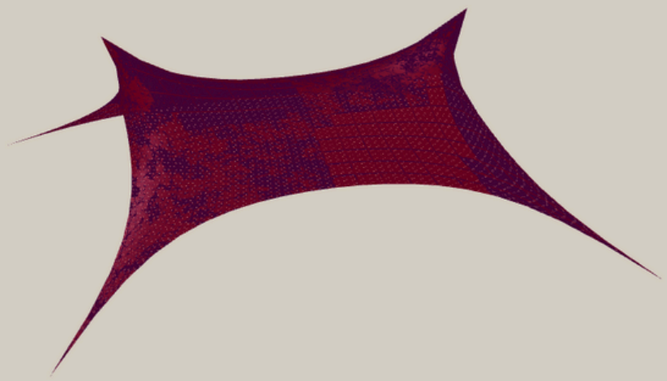
3. Specify the mesh edges of the surface and apply loads on it. Using the Anchor command will hold the edges in place while the rest of the structures simulate using the Solver command.



4. Make 2 points on top of the structure. Use the Closest Point command to find the closest points to the created ones and connect an Anchor to keep them from the simulation.



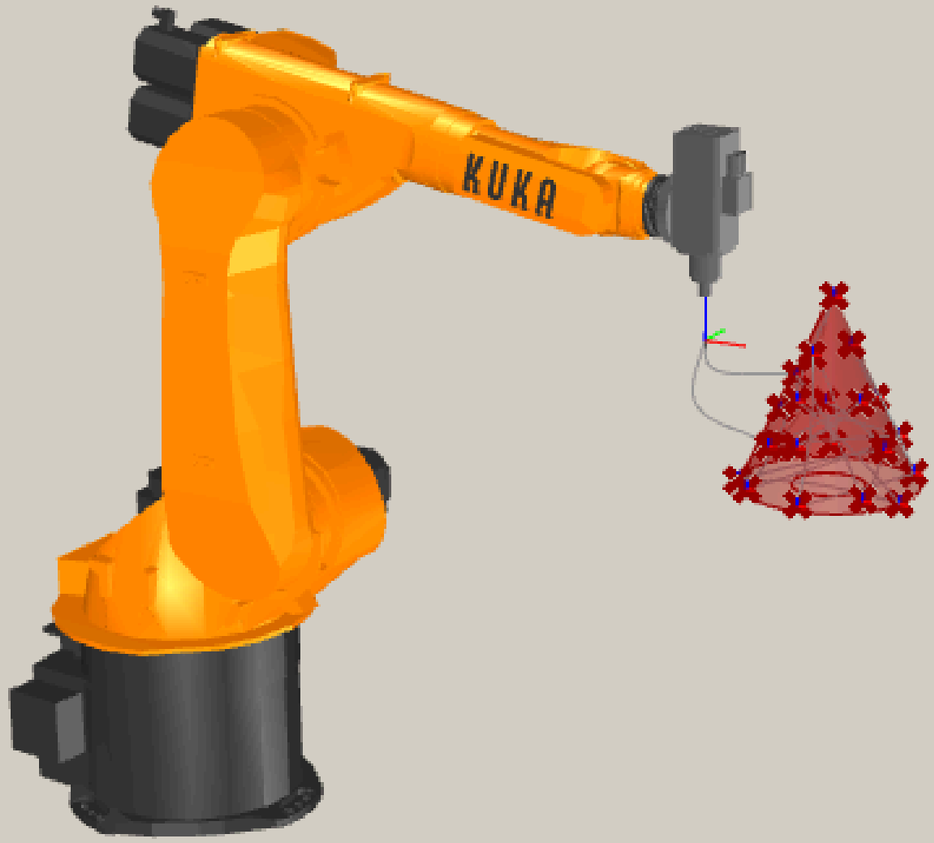
5. Create a mesh on the model and use the Explode Tree command to extract all the points from it.

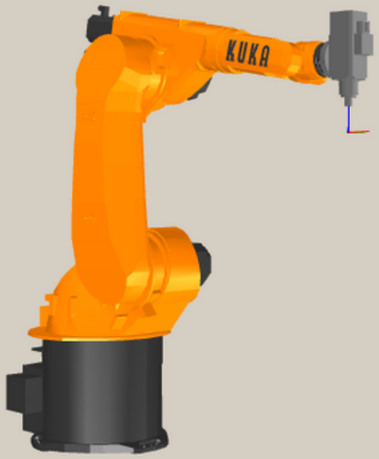


6. Use Weaverbird's Split Triangle Subdivision command to split the mesh into triangles and construct a domain for all the mesh edges. Disjoint the mesh into different pieces and use the Gradient command to colour them.

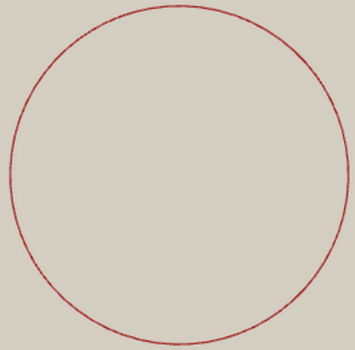
5.1 KUKA robot simulation

This model has been created using KUKA|prc, which is a plugin for Grasshopper. This is a parametric robot control plugin designed exclusively for KUKA robots. It supports programming and full kinematic simulation. KUKA|prc is developed to make industrial robots accessible to the creative field. It allows for fast prototyping and visualisation of robotic behaviours directly within Grasshopper.

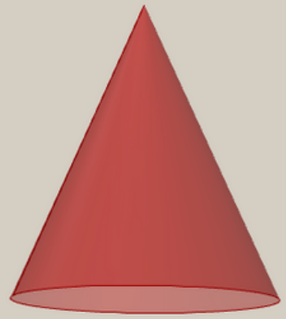




1. To start creating the model, first put the virtual robot on Grasshopper, which can be found on KUKA|prc section.



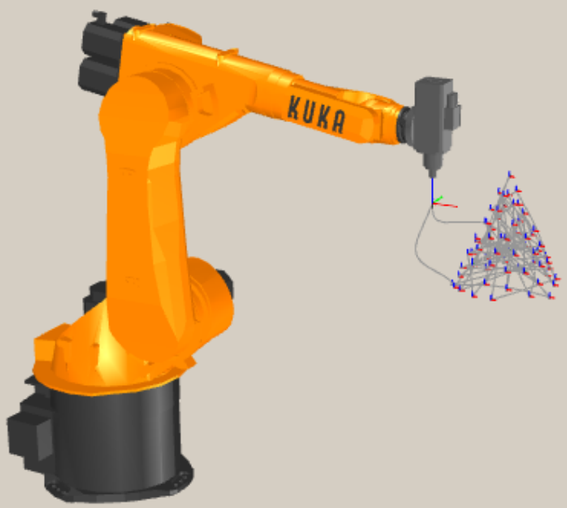
2. To start creating a model using the robot, start with a simple circle shape.



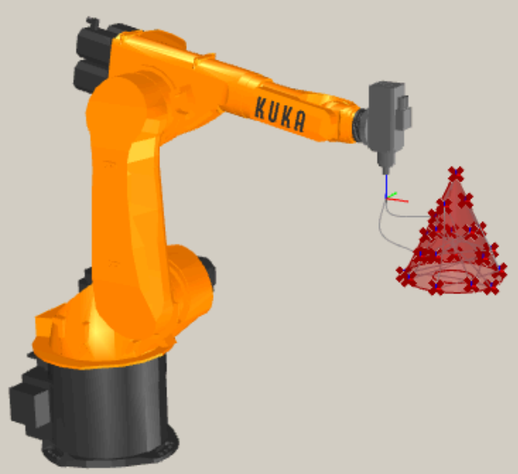
3. Then from the circle, create a cone, which will be the object being simulated using the robot.



4. To map out the path that the robot will follow to create this shape, I use the Populate Geometry command to create a path made out form points on the object.



5. Finally, telling the robot to use a planar movement, it starts following the path to create the cone.



6. After following all the points on the path, the robots simulates the object

References

1. Mater, N. (2023). Weekly Workflow: Panelize Irregular Geometry with in Grasshopper with LunchBox - PROVING GROUND. [online] PROVING GROUND. Available at: https://provingground.io/2023/09/06/weekly-workflow-panelize-irregular-geometry-with-in-grasshopper-with-lunchbox/ [Accessed 13 Jan. 2025].

2. MeshCan. (2025). Grasshopper Voronoi. [online] Available at: https://parametrichouse.com/parametric/grasshopper-voronoi/ [Accessed 13 Jan. 2025].

3. Bespokegeometry.com. (2015). Mesh Subdivision: Loop and Catmull-Clark – bespoke geometry. [online] Available at: https://www.bespokegeometry.com/2015/01/29/mesh-subdivision-loop-and-catmull-clark/ [Accessed 13 Jan. 2025].

4. 16zu9 Blog. (2010). Metaballs in Rhino/Grasshopper. [online] Available at: https://blog16zu9.wordpress.com/2010/04/29/metaballs-in-rhinograsshopper/ [Accessed 13 Jan. 2025].

5. KUKA|prc - parametric robot control for Grasshopper. [online] Food4Rhino. Available at: https://www.food4rhino.com/en/app/kukaprc-parametric-robot-control-grasshopper [Accessed 13 Jan. 2025].

6. Saili Sawantt (2022). Kangaroo 3D: A Form-finding, optimized, and constraint-solving engine. [online] Parametric Architecture. Available at: https://parametric-architecture.com/kangaroo-3d-a-form-finding-optimized-and-constraint-solving-engine/ [Accessed 13 Jan. 2025].

FIONA TOPORE

An MEng Architectural Engineering student from Tirana, Albania. Eager to learn more about and be a part of the Engineering industry. Hold a great interest in sustainable approaches and being able to create a better environment for the community in the future. Passion for hand drawing sketching and model making. Fluent in Albanian and English. Understand Italian and French (limited working proficiency)

Education

Sep 2021 - Present

MEng Architectural Engineering
Cardiff University, Cardiff

- Predicted: 2:1
- Third Year results:-
- Dissertation– 70%-
- Integrated building Design- 69%
- Tensile Structures – 73%

Sep 2019 - Jun 2021

A Level
Abbey College, Manchester

- A Levels: Mathematics A, Physics B, Art B

Sep 2018 - Jun 2019

A Level
Kings College, Tirana, Albania

- A Level: Mathematics A, Physics B, Chemistry B

Sep 2017 - Jun 2018

GCSE
New York Highschool, Tirana, Albania

- GCSEs: 8 x A*- A, including Maths, English, and Science

Work experience

Jul 2024 - Aug 2024

Intern
Whitby Wood, London

- Gained a lot of experience from working in the office. Helped the other engineers with stage 3/4 projects focusing on the Greenwich District, while working with the other interns on our own project, which we presented at the end of the internship.

Jan 2024 - Feb 2024

Bartender
The Philarmonic, Cardiff

- Worked a part-time job as a bartender alongside my studies.
- Learned to work well under pressure and be efficient in a fast-paced environment.

Jun 2023 - Jul 2023

Intern
Arkimade, Tirana, Albania

- Completed a CAD and Revit course over the summer, creating models of existing work from the studio (Residental) Became proficient in AutoCAD. Improved
- my communication and teamwork skills.
-

PROFESSIONAL MEMBERSHIP

- Student Member, ICE (Institution of Civil Engineers)
- Student Member, IStructE (Institution of Structural Engineers)
- Mentee, Student Mentor Scheme, Cardiff University

Technical skills

- Hand drawing, model making
- LUSAB;toCAD, MATLAB, Fusoin360, and Revit for university projects (Hotel Design, Student Hub, Ceremony Venue)
- Time Management, able to achieve and complete su icient work on assigned deadlines.
- Proficient in the use of Word, Excel, PowerPoint, and Adobe Photoshop.

ACHIEVEMENTS AND CERTIFICATES

- Chemistry Olympiad/Winner, The Ministry of Education, Albania, 2017
- Art Certificate/Winner, ISA, Manchester, 2020
- Ski Cup, Santa Christina, 2012

Interests

- Sports. Have been part of the swim team during my school years. Part of the Cardiff University Volleyball Club. Enjoy skiing.
- Art, drawing, model making, brain games, and cooking.

Thank you !